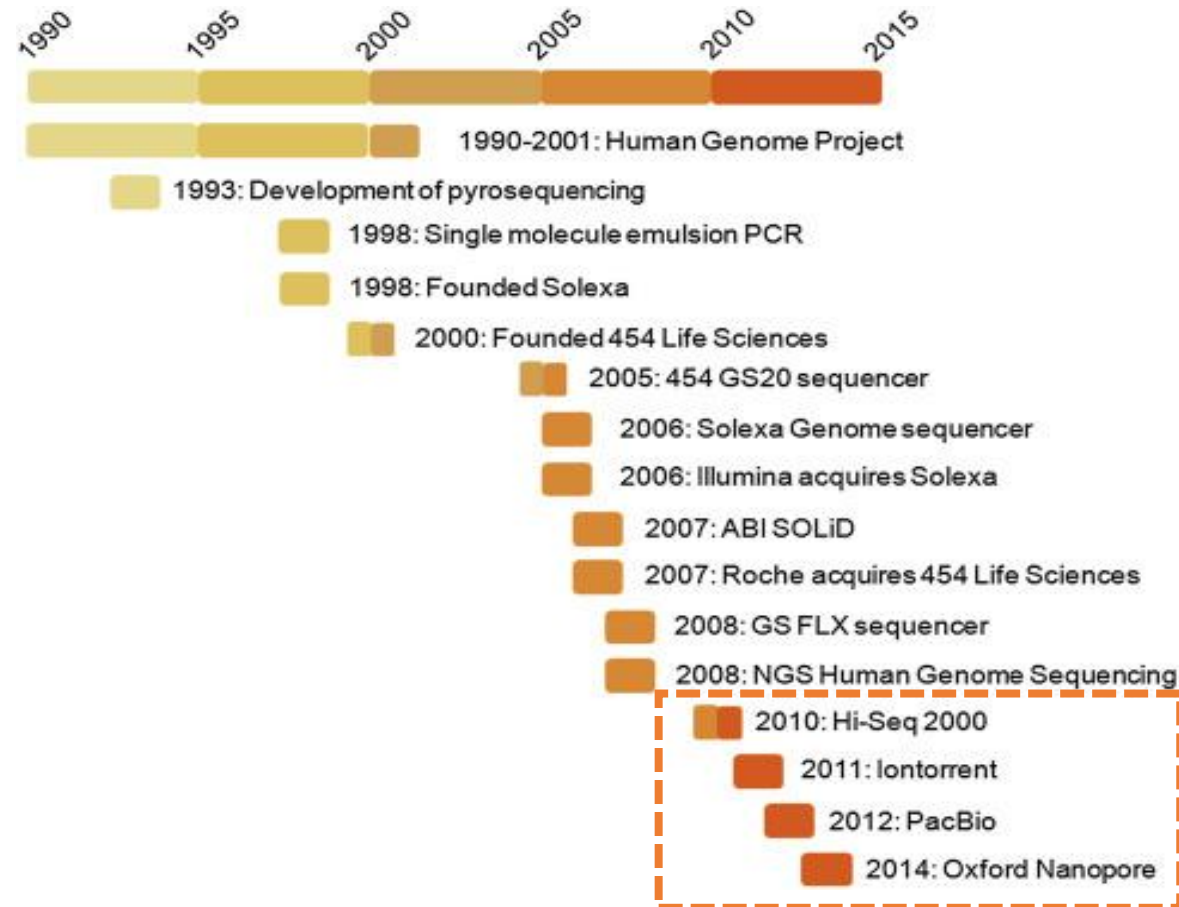


# Clase I. Sequencing Technologies, file formats, and genomic repositories

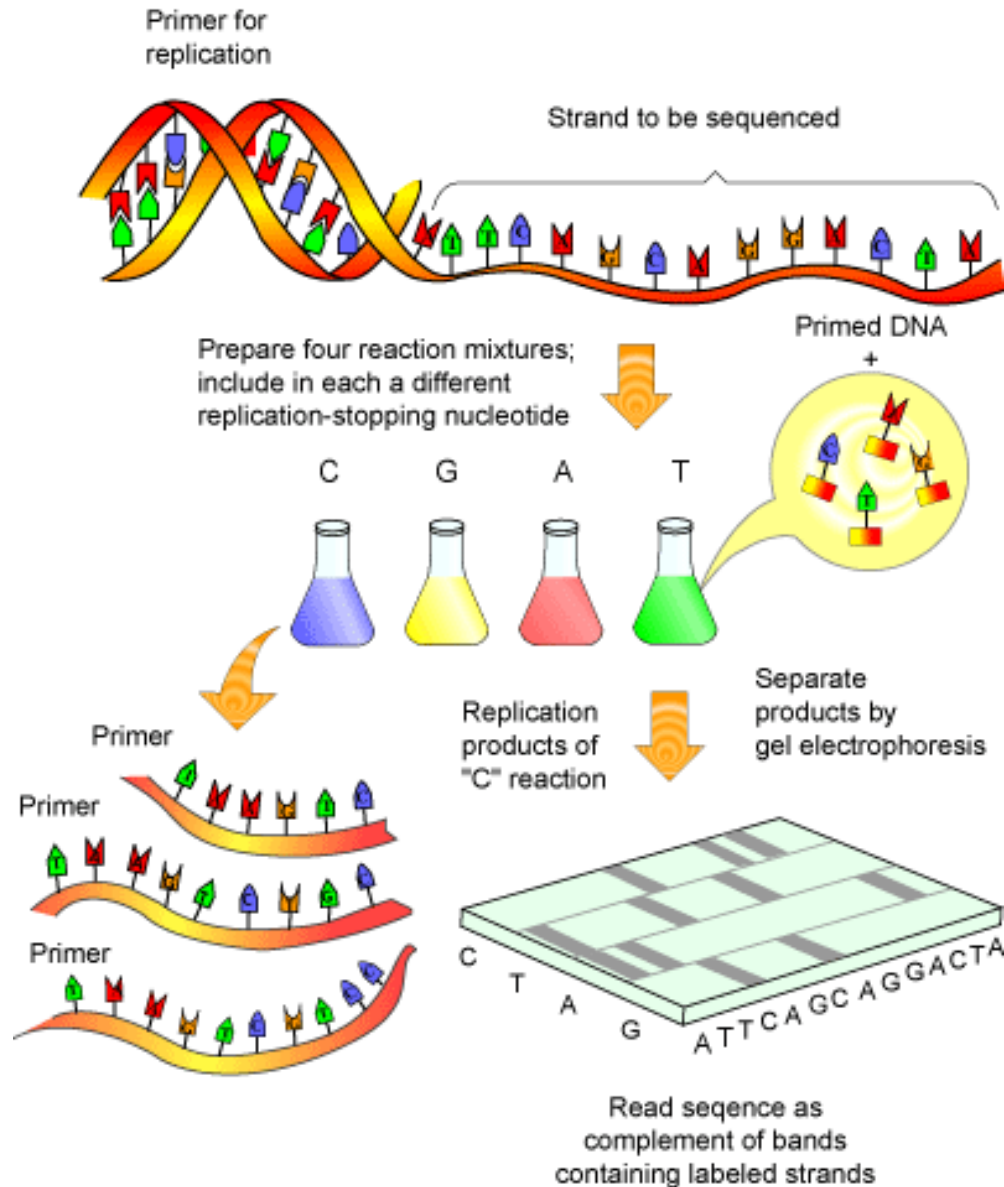


# Evolution of nucleotide sequencing

3 sequencing generations

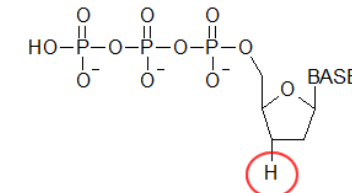
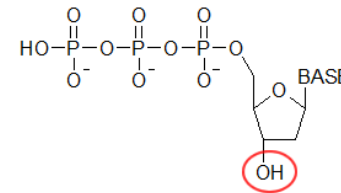


# Sanger sequencing or Terminator method



Based on PCR amplification → It is known as the synthesis method

Uses dideoxynucleotide triphosphate (ddNTPs) lacking 3'-OH required for phosphodiester bond



In the beginning, nucleotides were radio-labeled ( $^{35}\text{s}$  dCTP)

To identify the last nucleotide incorporated, 4 reactions were needed, and fragments were separated by an acrylamide gel with different nucleotide resolutions depending on the concentration.



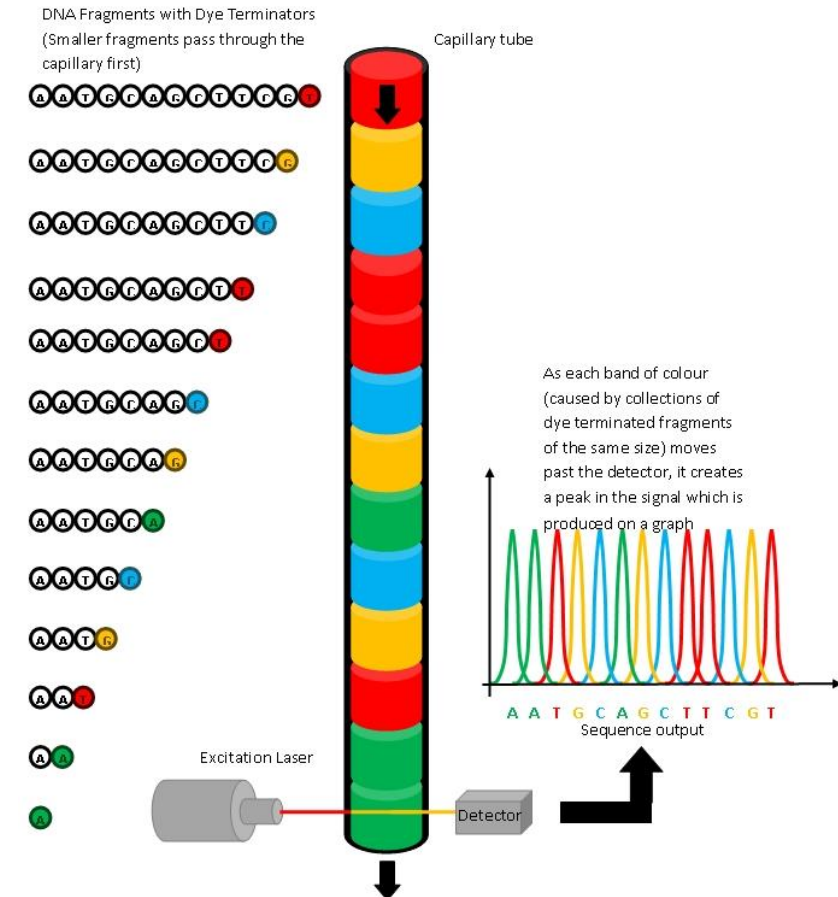
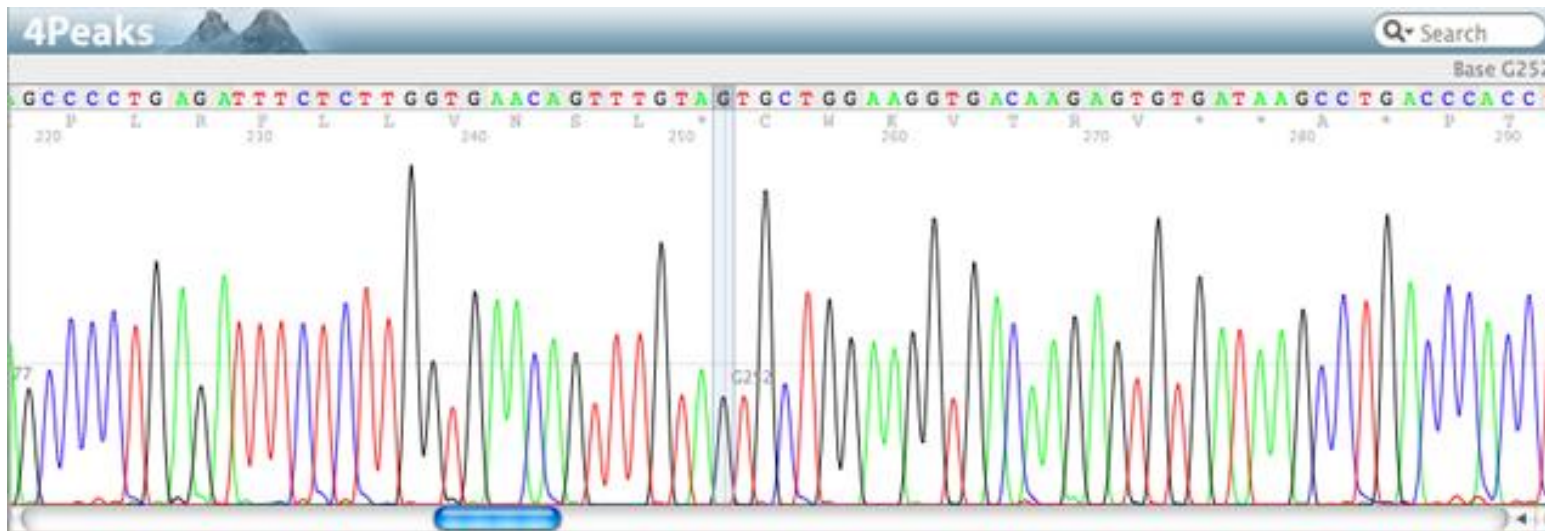


# Automatized sanger sequencing with fluorescent terminators (current)

A different fluorescence to each terminator allows for running just one reaction.

Fragments separation perform on capillary electrophoresis coupled to an electronic detector

The detector is connected to a computer and the signal is read on a chromatogram (electropherogram)



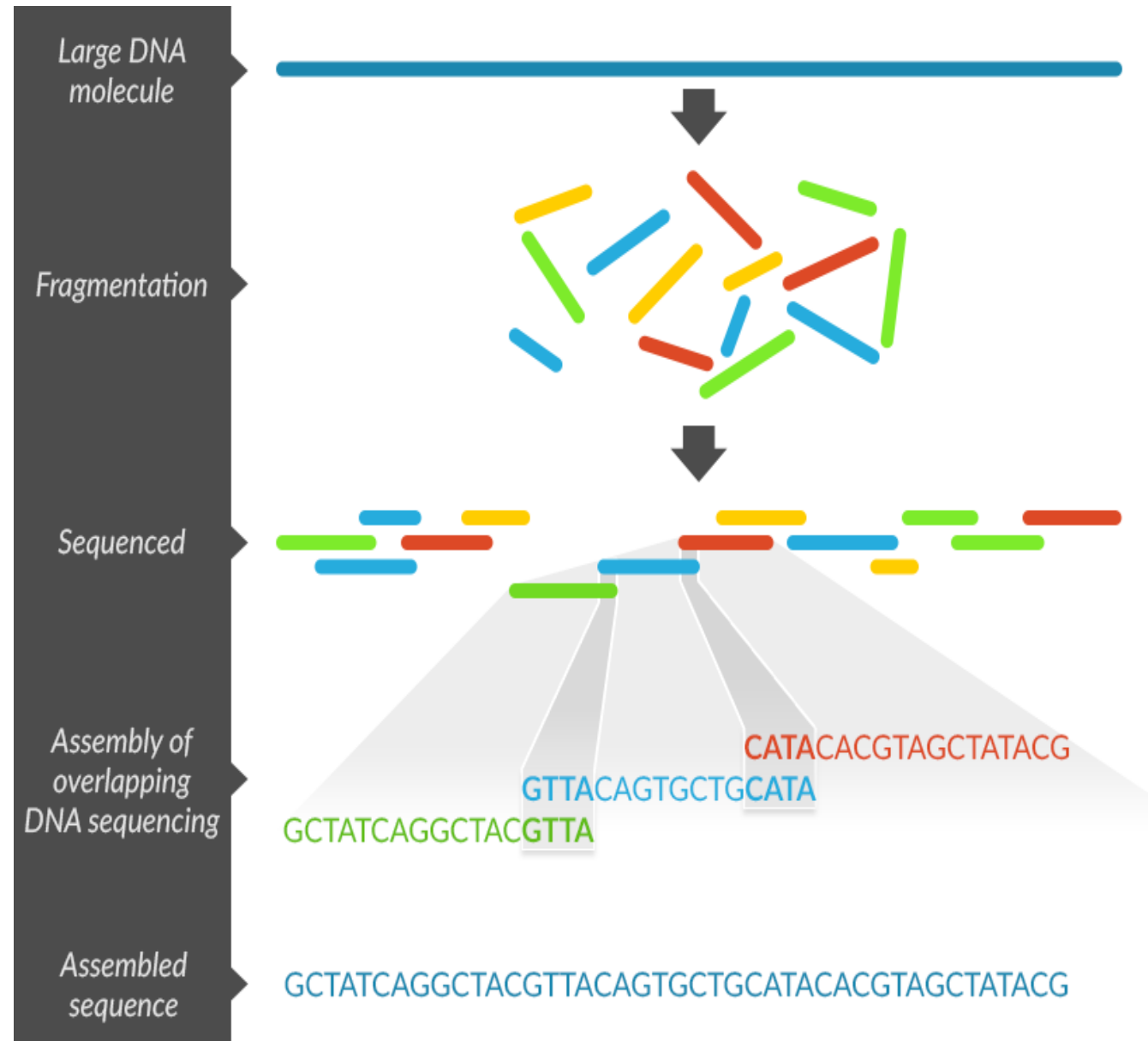
# Shotgun sanger sequencing

Sequencing Range: 100-1000bp

It was implemented to sequence long DNA fragments

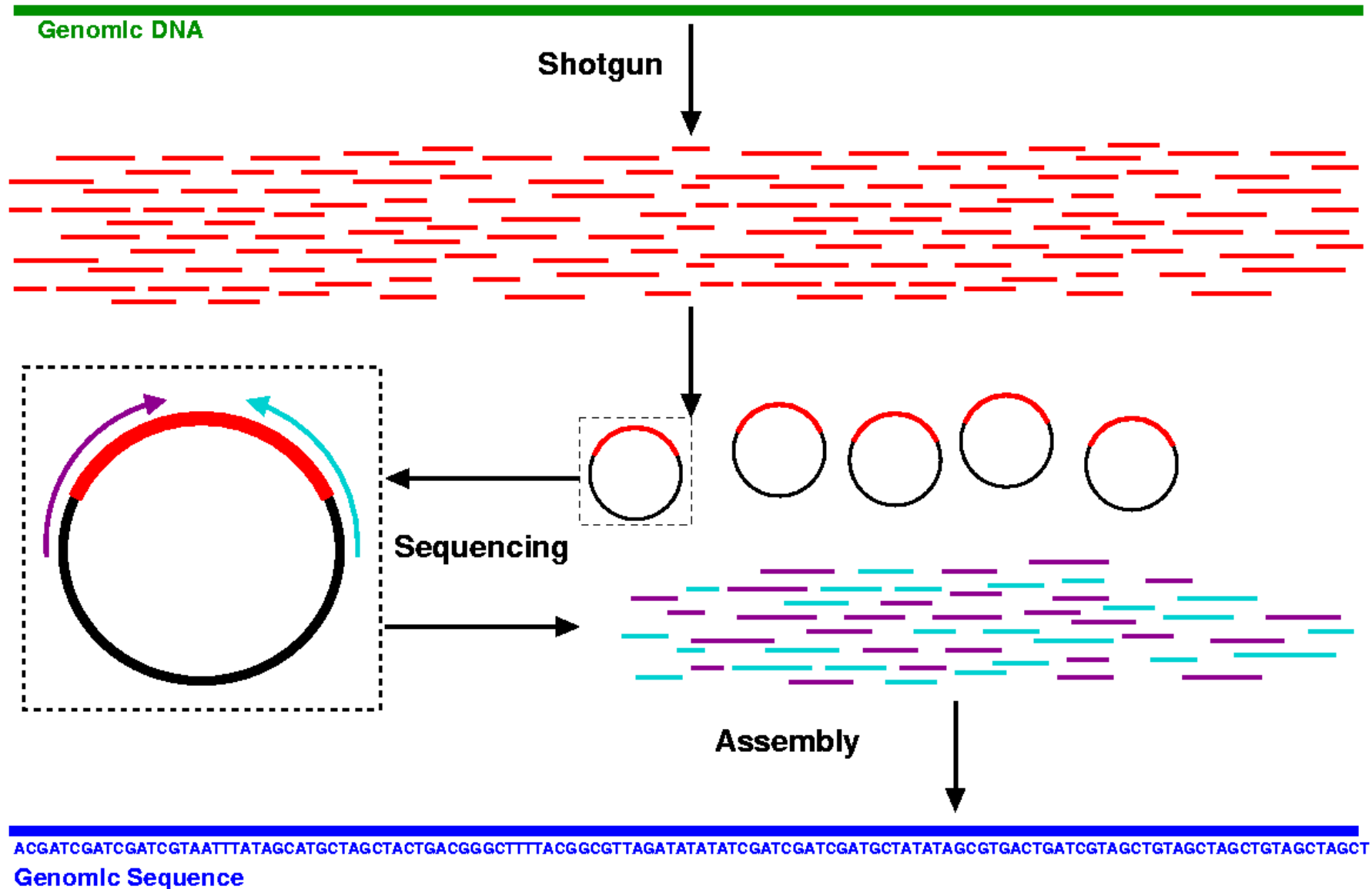
The principle is based on random fragmentation of the long DNA fragment.

Those fragments are sanger sequenced and posteriorly assembled based on overlapping sections to reconstruct the original DNA molecule.



# Whole-Genome shotgun sanger sequencing

Shotgun sequencing allows the start of whole-genome sequencing



# Key Sequencing Concepts



There are some concepts/features of sequencing that must to be defines before we continue

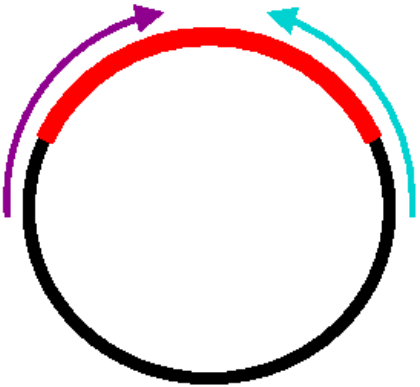


- > What do we sequence?
- > Sequencing one or both ends of the fragment
- > Fragment Inner distance
- > Sequencing Depth and Coverage

# What do we sequence?

A piece of DNA with two adapters sequences attached, one at each end, which are used for the sequencing reaction

On shotgun sequencing, the known regions around the insert will play as adapters



On next-generation sequencing, this sequences must be compatible with the sequencing technology





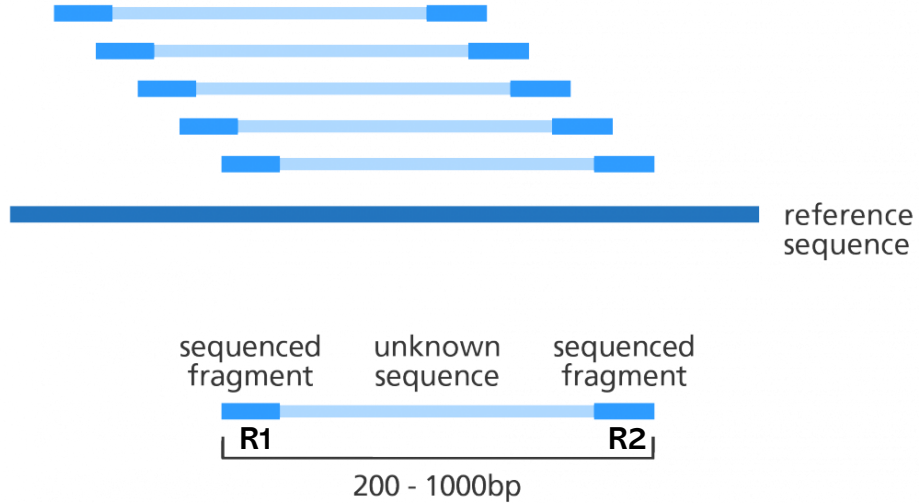
# Single-end or Paired-end sequencing

## Single-end reads



- **Single-end (SE):** sequencing of only one end of the fragment. One file is generated by the sequencer

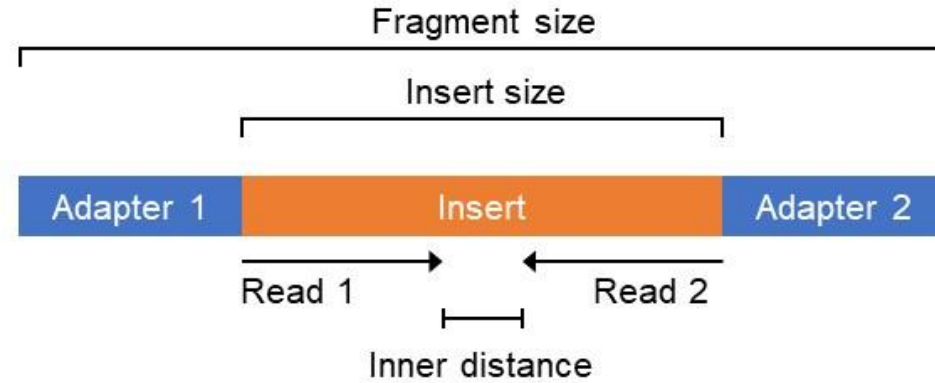
## Paired-end reads



- **Paired-end (PE):** sequencing of both ends of the fragment. This type of sequencing allows a better reconstruction of the sequenced fragment as we know the size of the fragment. Two files are generated by the sequencer (read 1 and read 2)

# Paired-end sequencing → Inner distance

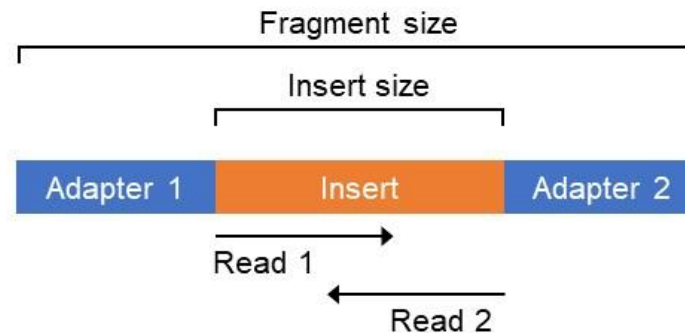
When sequencing paired-end, we will get one sequence by end of the fragment



Fragment size is the size of the DNA of interest plus the size of the adapters

Insert size is the size of the DNA fragment between the adapters, this size depends on the library preparation protocol

The Inner distance is the size of the DNA of interest that has not been covered by the sequencing procedure. Depending on the size of our reads we might or might not have inner distance



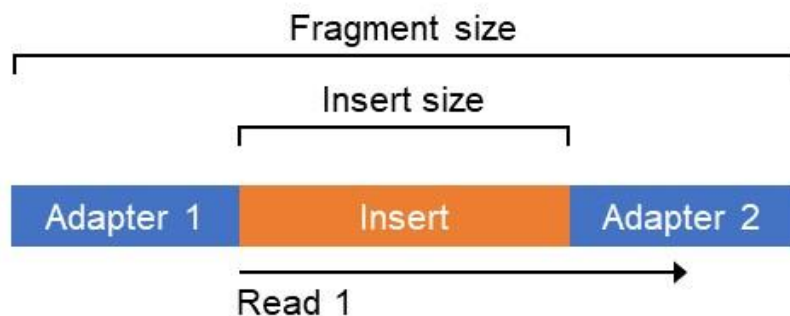
# Inner distance exercises

A. If I have an insert size of 350bp and I perform a 100nt PE sequencing, what is the length of my inner distance?

B. If I have an insert size of 500bp and I perform a 200nt SE sequencing, what is the length of my inner distance?

C. If I have an insert size of 250bp and I perform a 150nt PE sequencing, what is the length of my inner distance?

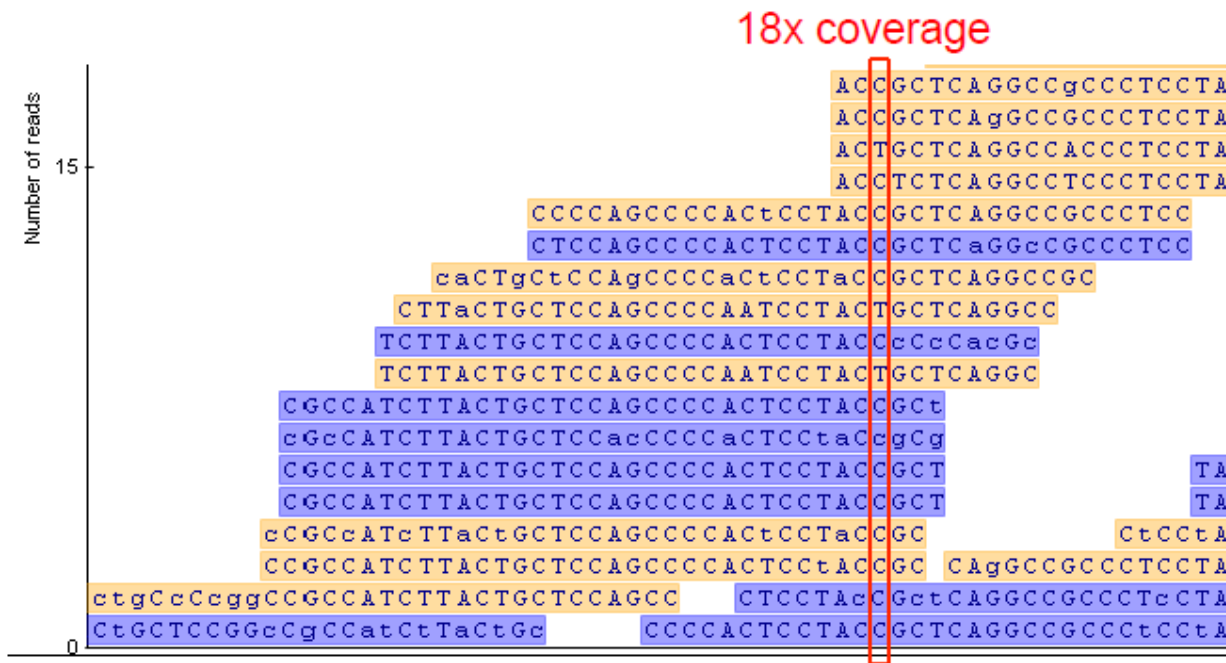
D. If I have an insert size of 100bp and I perform a 150nt PE sequencing, What Will I get as result?



# Sequencing Depth and Read Coverage

**Sequencing depth** is the number of reads (so, DNA fragments) sequenced. This number must be defined before sequencing according to the transcriptome/genome size. Organisms with big genomes will require more sequenced reads than organisms with small genomes.

**Read Coverage** is the number of sequences covering a specific region of the genome or transcriptome.



partly overlapping sequencing reads result from the multiple templates being sequenced across the flow cell

**Read Coverage** is much easier to calculate for **genome sequencing** as it is recommended 100x coverage to facilitate the genome assembly. Lander/waterman equation is used to estimate the number of reads need:

$$C = LN/G$$

C stand for coverage

G is the haploid genome length

L is the read length

N is the number of reads

What is the number of reads required to sequence the genome of zebrafish with coverage of 100x (Zebrafish haploid genome size  $1.7 \times 10^9$ )???

# Read Coverage for RNAseq

**Read Coverage** for RNA-seq is often calculated in terms of sequence depth by sample and will depend on the objective of your experiment.

Recommendations for eukaryotes organisms:

- For quick snapshots of highly expressed genes, 25 millions reads by samples is enough
- For a global view of gene expression and some information on alternative splicing, typically 60 million reads by sample will work (most of the published works are using this sequence depths)
- In-depth transcriptome exploration will need a minimum of 200 million reads is need
- Targeted sequencing 3 millions

Read length recommendations:

- mRNA profiling → SE 75bp
- Transcriptome assembly → PE 75bp or 100bp
- Small RNA → SE 50bp

OPEN ACCESS PEER-REVIEWED

RESEARCH ARTICLE

## Evaluating the Impact of Sequencing Depth on Transcriptome Profiling in Human Adipose

Yichuan Liu, Jane F. Ferguson, Chenyi Xue, Ian M. Silverman, Brian Gregory, Muredach P. Reilly, Mingyao Li

Published: June 24, 2013 • <https://doi.org/10.1371/journal.pone.0066883>

| Article | Authors | Metrics | Comments | Media Coverage |
|---------|---------|---------|----------|----------------|
| ⌵       |         |         |          |                |

Abstract

Introduction

Materials and Methods

Results

Discussion

Supporting Information

Author Contributions

References

Reader Comments

Figures

### Abstract

Recent advances in RNA sequencing (RNA-Seq) have enabled the discovery of novel transcriptomic variations that are not possible with traditional microarray-based methods. Tissue and cell specific transcriptome changes during pathophysiological stress in disease cases versus controls and in response to therapies are of particular interest to investigators studying cardiometabolic diseases. Thus, knowledge on the relationships between sequencing depth and detection of transcriptomic variation is needed for designing RNA-Seq experiments and for interpreting results of analyses. Using deeply sequenced Illumina HiSeq 2000 101 bp paired-end RNA-Seq data derived from adipose of a healthy individual before and after systemic administration of endotoxin (LPS), we investigated the sequencing depths needed for studies of gene expression and alternative splicing (AS). In order to detect expressed genes and AS events, we found that ~100 to 150 million (M) filtered reads were needed. However, the requirement on sequencing depth for the detection of LPS modulated differential expression (DE) and differential alternative splicing (DAS) was much higher. To detect 80% of events, ~300 M filtered reads were needed for DE analysis whereas at least 400 M filtered reads were necessary for detecting DAS. Although the majority of expressed genes and AS events can be detected with modest sequencing depths (~100 M filtered reads), the estimated gene expression levels and exon/intron inclusion levels were less accurate. We report the first study that evaluates the relationship between RNA-Seq depth and the ability to detect DE and DAS in human adipose. Our results suggest that a much higher sequencing depth is needed to reliably identify DAS events than for DE genes.

Figures

# Next-Generation sequencing platforms – Second Generation

- Corresponds to massive sequencing platforms and it is characterized by high depth
- >100x cheaper and faster than sanger sequencing
- During the development of this technology appear several platforms, however, Illumina is the gold standard now.

| Instrument | Method                      | Read Length | Yield | Quality | Value |
|------------|-----------------------------|-------------|-------|---------|-------|
| Illumina   | synthesis + fluorescence    | 250         | ++++  | +++++   | ++++  |
| SOLID      | ligation + fluorescence     | 75          | ++++  | +++     | +++   |
| Roche 454  | non-term NTP + luminescence | 600         | +     | ++++    | ++    |



# Illumina sequencing

# Illumina Sequencers



| Sequencing System           | iSeq™   | MiniSeq™ | MiSeq® | NextSeq® | HiSeq®              | HiSeq® X                              | NovaSeq®                 |
|-----------------------------|---------|----------|--------|----------|---------------------|---------------------------------------|--------------------------|
|                             |         |          |        |          | 4000                | Five/Ten                              | 6000                     |
| Output per run              | 1.2 Gb  | 7.5 Gb   | 15 Gb  | 120 Gb   | 1.5 Tb              | 1.8 Tb                                | 1 Tb - 6 Tb <sup>1</sup> |
| Instrument price            | \$19.9K | \$49.5K  | \$99K  | \$275K   | \$900K              | \$6M <sup>2</sup> /\$10M <sup>2</sup> | \$985K                   |
| Installed base <sup>3</sup> | NA      | ~600     | ~6,000 | ~2,400   | ~2,300 <sup>4</sup> |                                       | ~285                     |

1. Output per run for the S1, S2 and S4 flow cells equal 1 Tb, 2 Tb and 6 Tb, respectively assuming two flow cells per run
2. Based on purchase of 5 and 10 units for HiSeq X Five and HiSeq X Ten, respectively
3. Based on end of fiscal year 2017
4. Combined HiSeq family

I see you are getting sleepy, so....10 min coffee, cigarette, or any legal drug break!



# Multiplexing libraries

- During library preparation, on the sequencing adaptor, we can include a short sequence called **Index** to tag the sample.
- Add index to the libraries to allow **pooling of a large number of libraries on a single run** of Illumina sequencing (multiplexing)
- Currently, Up to 96 samples can be multiplexed in one run. Here, the number of libraries pooled will depend on the **sequencer and the objective** of the experiment.

| organism                               | genome size | depth | read length | no of samples |         |        |
|----------------------------------------|-------------|-------|-------------|---------------|---------|--------|
|                                        |             |       |             | MiSeq         | NextSeq | HiSeq* |
| Staphylococcus aureus (MRSA)           | 2.8 Mb      | 100x  | 2x300       | 45            |         |        |
| Mycobacterium tuberculosis (TB)        | 4.4 Mb      | 100x  | 2x300       | 29            |         |        |
| E. coli                                | 4.6 Mb      | 100x  | 2x300       | 27            |         |        |
| <i>Plasmodium falciparum</i> (malaria) | 22.9 Mb     | 100x  | 2x300       | 5             |         |        |

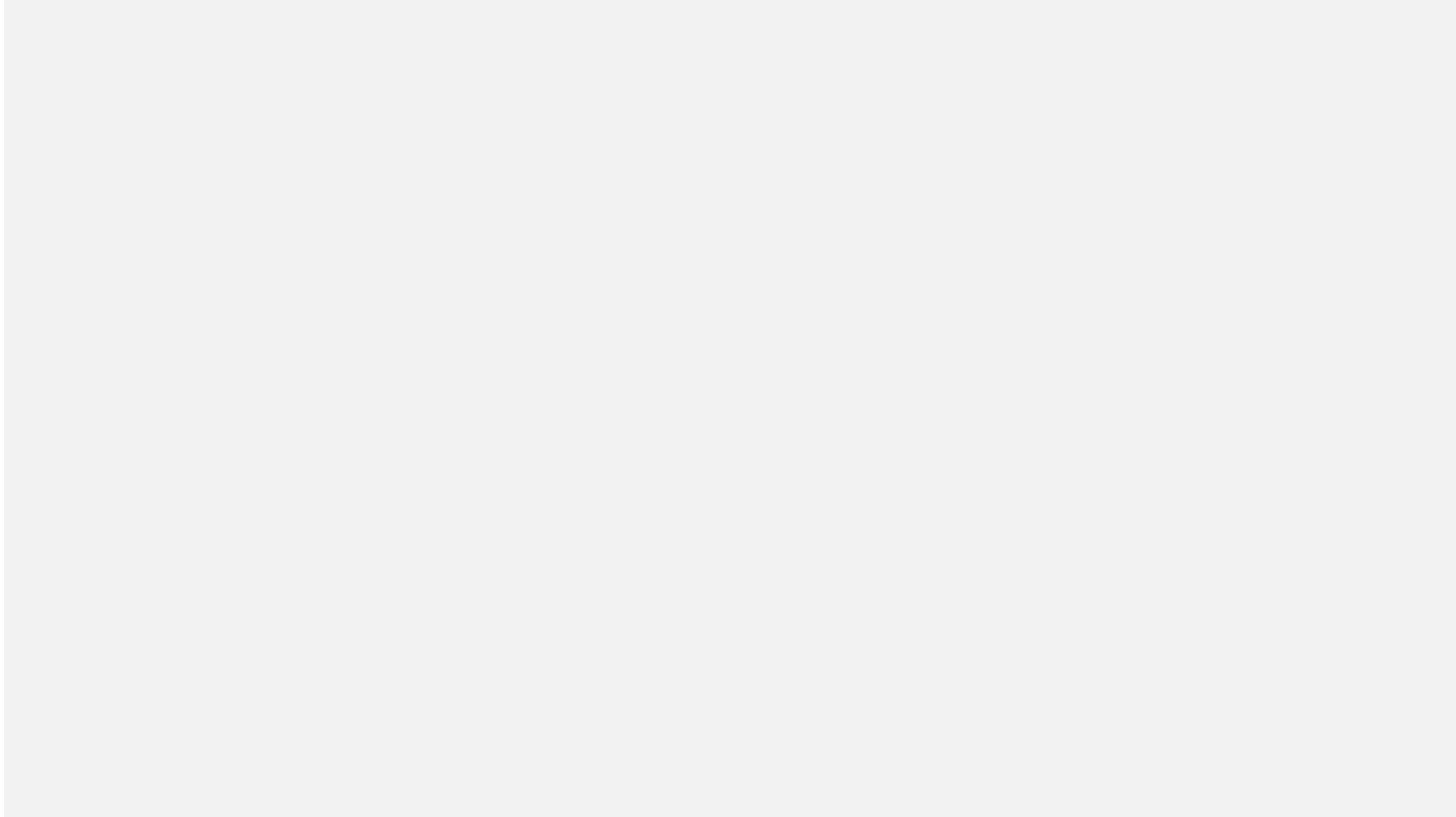
# Third Generation sequencing

- It was developed based on the need for larger and massive reads that help to resolve complex genomic structures such as **repetitive elements**, **copy number alterations**, **alternative splicing**, and **structural variations**.

|                                          | PacBio <sup>1</sup> |              | Oxford Nanopore <sup>2</sup>           |                            |
|------------------------------------------|---------------------|--------------|----------------------------------------|----------------------------|
| Instrument Specifications                | RS II (P6-C4)       | Sequel       | MinION                                 | PromethION                 |
| Average read length                      | 10 – 15 kb          | 10 – 15 kb   | Variable (up to 900 kb) <sup>3,4</sup> | *                          |
| Error rate                               | 10 – 15 %           | 10 – 15 %    | 5 – 15 % <sup>4,5</sup>                | *                          |
| Output                                   | 500 Mb – 1 Gb       | 5 Gb – 10 Gb | ~5 Gb <sup>4</sup>                     | *                          |
| # of reads                               | ~50k                | ~500k        | Variable (up to 1M) <sup>6,7</sup>     | *                          |
| Instrument price/Access fee <sup>a</sup> | \$700k              | \$350k       | \$1000 <sup>8</sup>                    | \$135k bundle <sup>9</sup> |
| Run price                                | ~\$400              | ~\$850       | \$500-\$900 <sup>7</sup>               | *                          |

# Thrid Generation — PacBio Platform

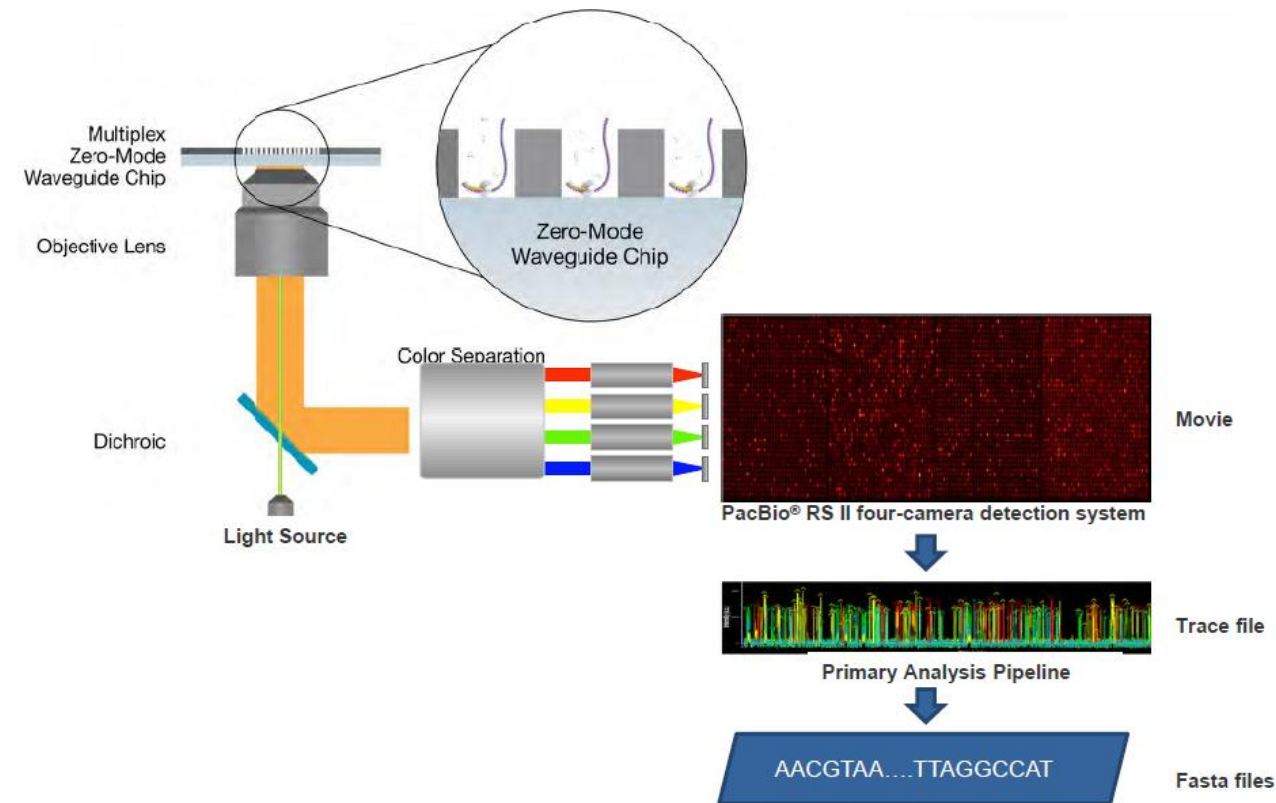
- The third-generation sequencing is also known as single molecule real time (SMRT) sequencing. In the case of PacBio technology, it is based on the use of a zero-mode waveguides (ZMWs)



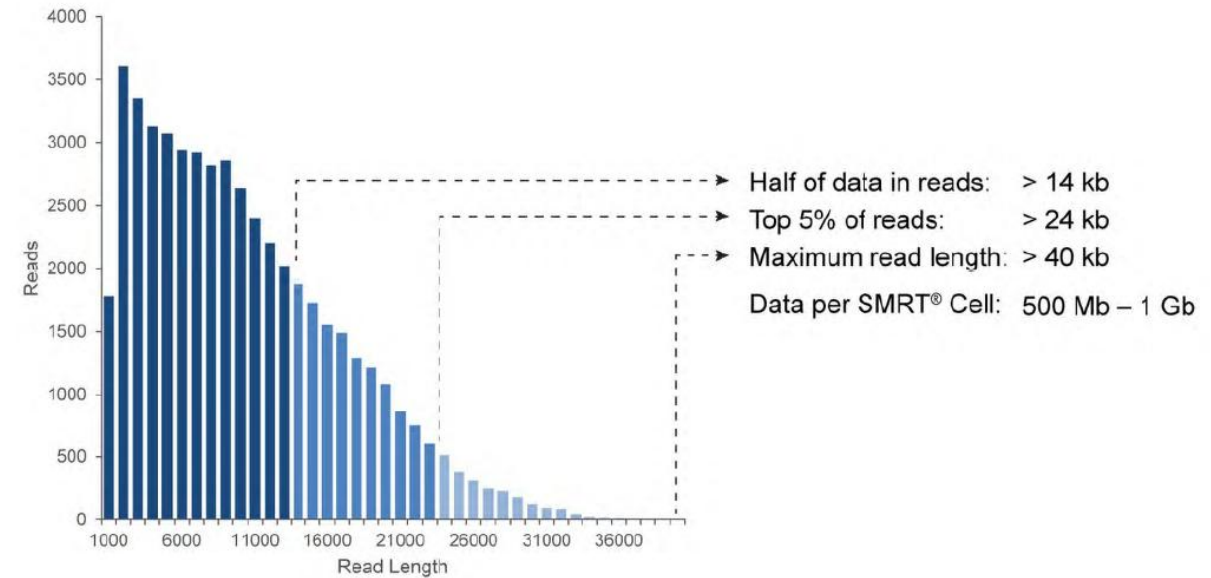
# Features of PacBio Sequencing

- Long reads can cover small genomes
- The data analysis does not require alignment, cDNA are sequenced as complete molecules
- It is possible to identify base modifications

## Detection system is optical



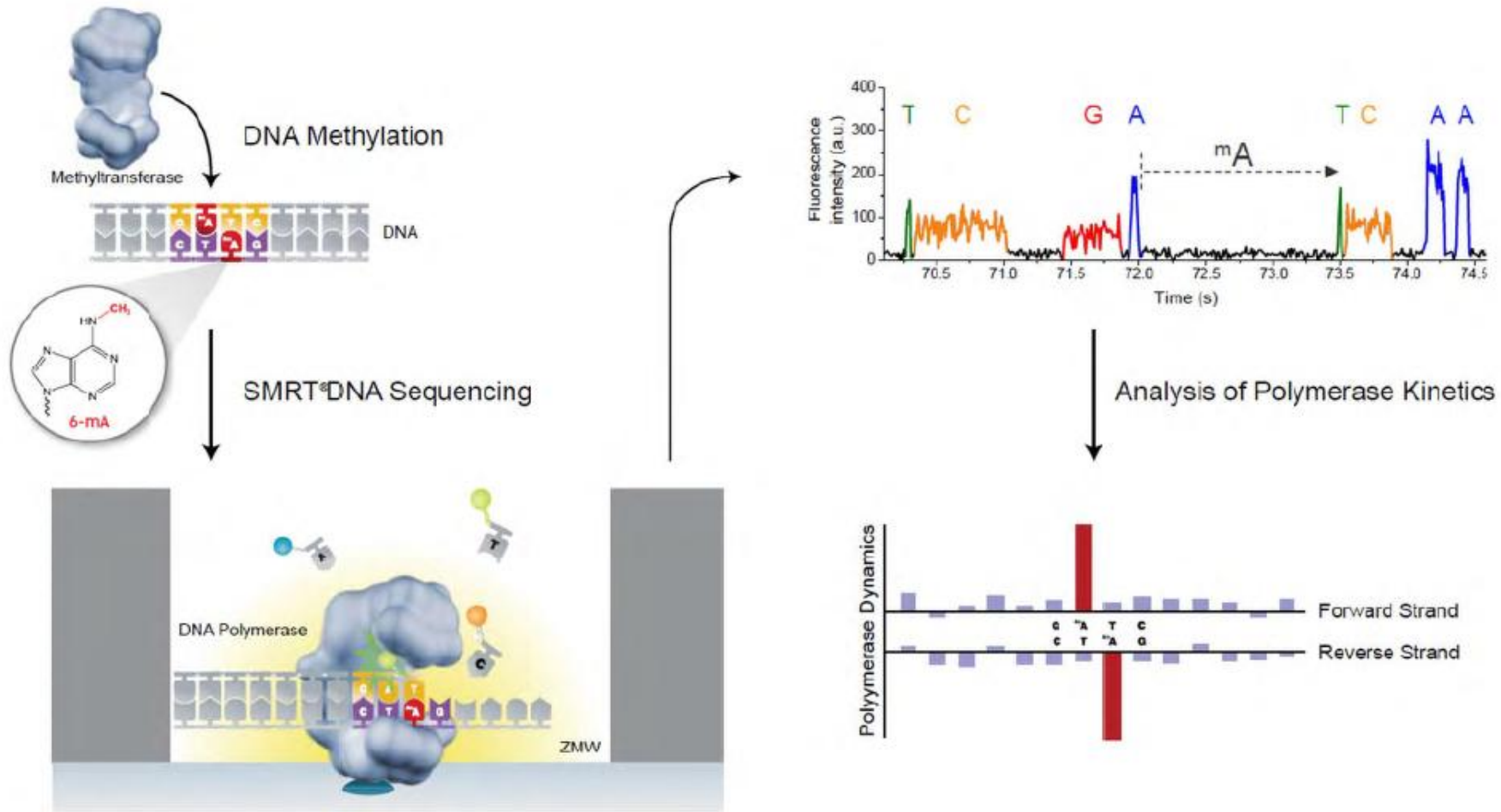
## Maximum length can be > 40kb



P6-C4, 4-hr movie, 20-kb BluePippin™ size-selected *E. coli* library (1 SMRT Cell)



# Base modification using kinetics of polymerization



Detect base modifications using the kinetics of the polymerization reaction during normal sequencing

# High precision using CCS (99,9999%, QV60)

## PacBio Evaluation Study for Microbial Genomes



**Table 3 Genome assembly continuity and correctness comparison to secondary technologies**

| Organism                   | Assembled with                           | Assembly bp    | Contigs  | N50            | LAP                | Discordant Bases | QV            |
|----------------------------|------------------------------------------|----------------|----------|----------------|--------------------|------------------|---------------|
| <i>E. coli</i> K12         | MiSeq 100X 2x150bp 300bp (MaSuRCA iCORN) | 4682345        | 139      | 113852         | 9.68E + 07         | 28               | 52.23         |
|                            | 454 50X                                  | 4569757        | 93       | 117490         | -9.73E + 07        | 17               | 54.29         |
|                            | <b>PBcR 200X</b>                         | <b>4653486</b> | <b>1</b> | <b>4653486</b> | <b>-9.64E + 07</b> | <b>3</b>         | <b>&gt;60</b> |
| <i>E. coli</i> O157:H7     | MiSeq 100X 2x150bp 500bp (SPAdes iCORN)  | 5433737        | 413      | 133641         | -3.67E + 07        | 62               | 49.43         |
|                            | 454 22X + 8X 5Kbp + 10X 10Kbp            | 5347050        | 409      | 133665         | -3.73E + 07        | 66               | 49.09         |
|                            | <b>PBcR 200X</b>                         | <b>5611389</b> | <b>9</b> | <b>4324437</b> | <b>-3.66E + 07</b> | <b>0</b>         | <b>&gt;60</b> |
| <i>B. trehalosi</i>        | MiSeq 100X 2x150bp 500bp (SPAdes iCORN)  | 2377594        | 83       | 222446         | -3.31E + 07        | 10               | 53.76         |
|                            | 454 50X                                  | 2364704        | 66       | 117742         | -3.32E + 07        | 9                | 54.20         |
|                            | <b>PBcR 200X</b>                         | <b>2411068</b> | <b>1</b> | <b>2411068</b> | <b>-3.27E + 07</b> | <b>0</b>         | <b>&gt;60</b> |
| <i>M. haemolytica</i>      | MiSeq 100X 2x150bp 500bp (MaSuRCA iCORN) | 2721965        | 89       | 84094          | -3.33E + 07        | 47               | 47.63         |
|                            | <b>PBcR 200X</b>                         | <b>2736037</b> | <b>1</b> | <b>2736037</b> | <b>-3.31E + 07</b> | <b>0</b>         | <b>&gt;60</b> |
| <i>F. tularensis</i>       | MiSeq 100X 2x250bp 500bp (SPAdes iCORN)  | 1825374        | 130      | 24065          | -1.33E + 07        | 0                | >60           |
|                            | 454 50X                                  | 1655657        | 326      | 7316           | -1.33E + 07        | 28               | 47.72         |
|                            | <b>PBcR 300X</b>                         | <b>1877407</b> | <b>3</b> | <b>573021</b>  | <b>-1.33E + 07</b> | <b>0</b>         | <b>&gt;60</b> |
| <i>S. enterica</i> Newport | MiSeq 56X 2x150bp 500bp (MaSuRCA iCORN)  | 5187269        | 114      | 195780         | -2.24E + 07        | 360              | 41.59         |
|                            | 454 23X + 2X 10Kbp                       | 5005089        | 172      | 372513         | -2.25E + 07        | 39               | 51.08         |
|                            | <b>PBcR 200X</b>                         | <b>5029197</b> | <b>2</b> | <b>4919684</b> | <b>-2.24E + 07</b> | <b>2</b>         | <b>&gt;60</b> |



WE GOT A BIG AMOUNT OF DATA...SO, WHAT NOW??



# Some well-know data repositories

- ENA / European Nucleotide Archive
- SRA /Sequence Read Archive
- GEO /Gene Expression Omnibus
- BioSD / BioSamples Database
- TIARA /Total Integrated Archive of short-Read and Array

CHECK COMPLETE LIST HERE: [HTTPS://WWW.NATURE.COM/SDATA/POLICIES/REPOSITORIES](https://www.nature.com/sdata/policies/repositories)

# Why do we care about repositories??



- ✓ Share raw data with the scientific community (Open source — Open science → transparency)
- ✓ Every good scientific journal will ask you to release your data before your accepted paper gets published
- ✓ Constitute a source of biological information that can be useful for other scientists around the globe.

# European Nucleotide Archive



Examples: [BN000065](#), [histone](#)

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## European Nucleotide Archive

The European Nucleotide Archive (ENA) provides a comprehensive record of the world's nucleotide sequencing information, covering raw sequencing data, sequence assembly information and functional annotation. [More about ENA](#)

Access to ENA data is provided through the browser, through search tools, large scale file download and through the API.

### Popular

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Examples: [BN000065](#), [histone](#)

Search

[Advanced search](#)

### Latest ENA news

**12 Jul 2017:** [Submission service maintenance - 14/7/17 to 17/7/17](#)

Webin submission services will not be available between Friday 14/7...

# European Nucleotide Archive

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## Study: PRJEB10137

RNAseq from mature ductal cells from nkx6.1:GFP zebrafish lines

View: [Project XML](#) [Study XML](#)

Download: [Project XML](#) [Study XML](#)

### Name

Transcriptome of mature ductal cells

### Submitting Centre

GIGA-R, University of Liege

### Secondary accession(s)

ERP011343

### Description

Background: In contrast to mammals, the zebrafish has the remarkable capacity to regenerate very efficiently its pancreatic beta cells. Understanding the mechanisms of regeneration in zebrafish and the differences with mammals will be fundamental to discovering molecules able to stimulate the regeneration process in mammals. To identify the pancreatic cells able to give rise to new beta cells in zebrafish, we generated new transgenic lines allowing the tracing of multipotent pancreatic progenitors and endocrine precursors. Results: Using novel bacterial artificial chromosome transgenic nkx6.1 and ascl1b reporter lines, we established that nkx6.1-positive cells give rise to all the pancreatic cell types and ascl1b-positive cells give rise to all the endocrine cell types in the zebrafish embryo. These two genes are initially co-expressed in the pancreatic primordium and their domains segregate, not as a result of mutual repression, but through the opposite effects of Notch signaling, maintaining nkx6.1 expression while repressing ascl1b in progenitors. In adult zebrafish, nkx6.1 expression persists exclusively in the ductal tree at the tip of which its expression coincides with Notch active signaling in centroacinar/terminal end duct cells. Tracing these cells reveals that they are able to differentiate into other ductal cells and into Insulin-expressing cells in normal – non-diabetic – animals. This capacity of ductal cells to generate endocrine cells is supported by the detection of ascl1b in the nkx6.1:GFP ductal cell transcriptome. This transcriptome also reveals, besides actors of the Notch and Wnt pathways, several novel markers such as id2a. Finally, we show that beta cell ablation in adult zebrafish triggers proliferation of ductal cells and their differentiation into Insulin-expressing cells. Conclusions: We have shown that, in the zebrafish embryo, nkx6.1+ cells are bona fide multipotent pancreatic progenitors, while ascl1b+ cells represent committed endocrine precursors. In contrast to mouse, pancreatic progenitor markers nkx6.1 and pdx1 continue to be expressed in adult ductal cells, a subset of which we show are still able to proliferate and undergo ductal and endocrine differentiation, providing the first robust evidence of the existence of pancreatic progenitor/stem cells in adult zebrafish. Our findings support the hypothesis that nkx6.1+ pancreatic progenitors contribute to beta cell regeneration. Further characterization of these cells will open up new perspectives for anti-diabetic therapies.

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|----------------------------|------------------------------|----------------------------|----------------------------|---------------------------|--------|-----------------|---------------------|----------------|--------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|------------------------|------------------------|------------------------|---------------------------|
| <a href="#">PRJEB10137</a> | <a href="#">SAMEA3498332</a> | <a href="#">ERS805481</a>  | <a href="#">ERX1054380</a> | <a href="#">ERR977397</a> | 7955   | Danio rerio     | Illumina HiSeq 2000 | PAIRED         | <a href="#">File 1</a><br><a href="#">File 2</a> | <a href="#">File 1</a><br><a href="#">File 2</a> | <a href="#">Fastq file 1</a><br><a href="#">Fastq file 2</a> | <a href="#">Fastq file 1</a><br><a href="#">Fastq file 2</a> | <a href="#">File 1</a> | <a href="#">File 1</a> |                        |                           |
| <a href="#">PRJEB10137</a> | <a href="#">SAMEA3498333</a> | <a href="#">ERS805482</a>  | <a href="#">ERX1054381</a> | <a href="#">ERR977398</a> | 7955   | Danio rerio     | Illumina HiSeq 2000 | PAIRED         | <a href="#">File 1</a><br><a href="#">File 2</a> | <a href="#">File 1</a><br><a href="#">File 2</a> | <a href="#">Fastq file 1</a><br><a href="#">Fastq file 2</a> | <a href="#">Fastq file 1</a><br><a href="#">Fastq file 2</a> | <a href="#">File 1</a> | <a href="#">File 1</a> |                        |                           |
| <a href="#">PRJEB10137</a> | <a href="#">SAMEA3498334</a> | <a href="#">ERS805483</a>  | <a href="#">ERX1054382</a> | <a href="#">ERR977399</a> | 7955   | Danio rerio     | Illumina HiSeq 2000 | PAIRED         | <a href="#">File 1</a><br><a href="#">File 2</a> | <a href="#">File 1</a><br><a href="#">File 2</a> | <a href="#">Fastq file 1</a><br><a href="#">Fastq file 2</a> | <a href="#">Fastq file 1</a><br><a href="#">Fastq file 2</a> | <a href="#">File 1</a> | <a href="#">File 1</a> |                        |                           |

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# Sequence Read Archive

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## SRA

Sequence Read Archive (SRA) makes biological sequence data available to the research community to enhance reproducibility and allow for new discoveries by comparing data sets. The SRA stores raw sequencing data and alignment information from high-throughput sequencing platforms, including Roche 454 GS System®, Illumina Genome Analyzer®, Applied Biosystems SOLiD System®, Helicos Heliscope®, Complete Genomics®, and Pacific Biosciences SMRT®.

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